Milestone 3 Report: Rendering & Signal Processing

Procedural Terrain Generator

This stage aims to enable the realistic image formation of the procedural terrain generator.
The aim is to improve the visualization aspect of generated terrain by applying lighting calculations, surface representation, and sampling methods.
The rendering pipeline in this project involves combining procedural terrain generation with height-based material assignment and Phong lighting, as well as Super Sampling-based antialiasing, to produce a realistic terrain image.

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1. **Shading Model**

lighting.py: Phong lighting
renderer.py: applies lighting to terrain
Phong Illumination Model
The terrain renderer applies the Phong shading model to simulate the interaction between light and the terrain surface. Phong shading was selected due to its balance between realism and performance.
Ambient Component
Ambient lighting is used to simulate global illumination and prevents completely dark areas.
Diffuse Component
Diffuse lighting ensures that light intensity is proportional to the angle between the terrain surface normal and the light direction. Surfaces facing the light are illuminated more.
Specular Component
Specular lighting is responsible for sharp, bright reflections on terrain surfaces and contributes to realistic texture detail.

2. Texture or Surface Representation

material.py
Height-Based Procedural Texturing
The terrain utilizes a procedural height-based surface representation without using external image textures. Instead, terrain height is used to assign different materials. This allows for automatic generation of natural terrain landscapes.
Height
Below -2
-2 - 0
0 - 3
3 - 8
8 - 13
Above 13
Surface
Deep Water
Shallow Water

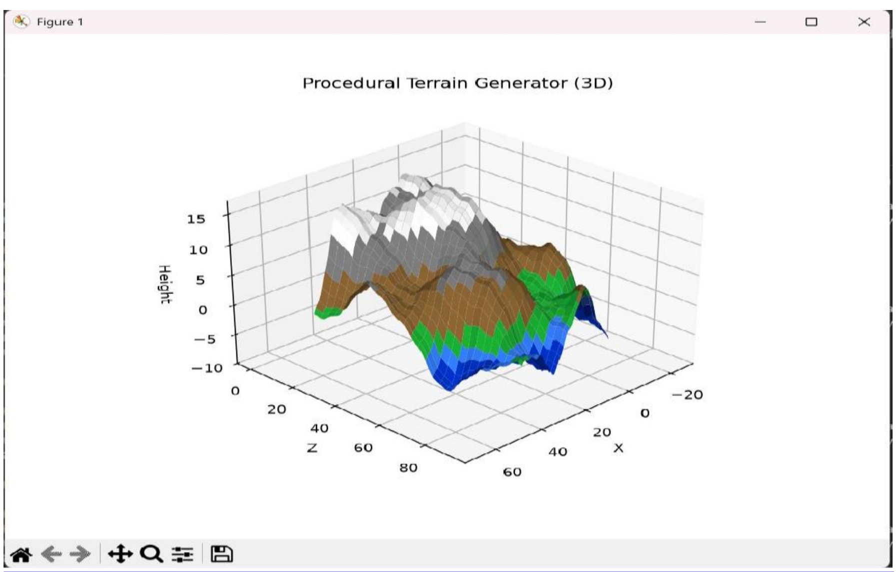
Grass
Dirt
Rock
Snow

3. Sampling and Antialiasing Method terrain_generator.py

Super sampling is used as an antialiasing technique to improve image quality. The method involves generating terrain at a higher resolution and then downsampling it to the final desired resolution.
1. Higher resolution terrain data is generated.
2. Multiple terrain points are sampled for each final terrain point.
3. Nearby samples are averaged when downsampling.
Super sampling minimizes aliasing effects such as jagged edges and blocky transitions, resulting in a smoother and more consistent terrain appearance.

Source Code

https://github.com/Lokaalee/Procedural-Terrain-Gene



4. Artifact Analysis

Aliasing

Aliasing arises from the discrete representation of continuous terrain features using an inadequate number of samples, resulting in staircased edges and abrupt surface changes. Super sampling reduces aliasing by oversampling the terrain and then averaging samples to achieve a smoother result.
Low resolution sampling

60 x 60 terrain samples

Possible aliasing
Super sampling

120 x 120 terrain samples

Average neighbouring samples

60 x 60 final terrain

Noise

Uncontrolled and irregular variations are considered noise. The generator reduces noise by using deterministic procedural functions and a fixed random seed (random.seed(42)).

Distortion

Distortion happens when the transformation from 3D terrain coordinates to 2D image causes a change in the shape of features. Consistent scaling and rotation transformations before visualization help preserve the spatial relationships.

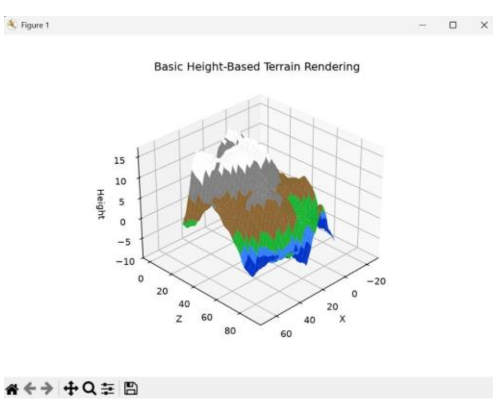
Original Terrain Coordinates
Scaling Transformation

Rotation Transformation

3D Visualization

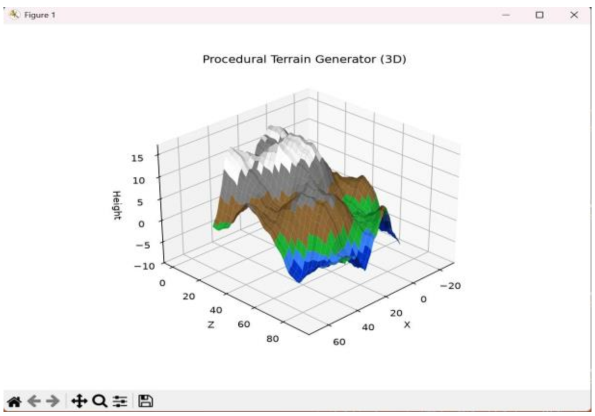
5. Comparison of Rendering Strategies

Strategy 1: Basic Height-Based Rendering
This method directly assigns colours based on height values without incorporating any lighting calculations.



Strategy 2: Phong Shading with Super Sampling

This project's chosen method incorporates material assignment, Phong lighting, and super sampling to produce more realistic results.
Feature



Feature	Basic height -based rendering	Phong shading+super sampling rendering
Lighting	No light model is applied. Terrain colours are displayed directly based on height values.	Uses the Phong illumination model with ambient, diffuse and specular lighting components to simulate light interaction.
Surface representation	Uses simple colour mapping where terrain elevation directly determines the displayed colour.	Uses procedural height-based materials where different elevation ranges represent water, grass, dirt, rock and snow
Realism	Low realism because terrain surfaces appear flat and do not respond to lighting conditions.	Higher realism due to realistic shading, depth perception, and varied surface materials.
Computation cost	Lower computation cost due to fewer calculations and no lighting processing.	Higher computation cost due to normal calculations and additional sampling.